

LINGUA AI

API Integration Reference

REST API specification for mobile client integration

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Base URL: <https://ailanguage.recipehubapi.com>
OpenAPI / Swagger: <https://ailanguage.recipehubapi.com/docs>
Classification: Internal — development team

Document scope

This reference defines the production REST contract between the Lingua AI mobile application and the backend service. It specifies authentication, the two supported conversation flows, every endpoint with request/response schemas, and integration rules the client must follow.

The server persists users, chats, and messages in MongoDB. The AI system prompt (systemInstruction) is composed once at chat creation and is never returned to or resent by the client.

Parameter	Value
Base URL	https://ailanguage.recipehubapi.com
API key header	x-api-key
Production key	KJSD84JIDNMZI83_3USLA983@8347SLDKJLE21D
Content-Type	application/json; charset=utf-8
Rate limit	300 requests / minute / IP

1. Authentication

All endpoints except GET /health and routes under /docs require a valid API key in the x-api-key request header.

x-api-key: KJSD84JIDNMZI83_3USLA983@8347SLDKJLE21D Content-Type: application/json; charset=utf-8

Status	Condition	Response
401	Missing or invalid key	{"error": "Unauthorized", "message": "Valid x-api-key header is required"}
400	Validation failure	{"error": "<message>", "details": {...}}
404	Resource not found	{"error": "Chat not found"}
500	Server error	{"error": "Internal server error"}

2. Client identity

Every chat operation requires deviceId (anonymous device) and/or userId (authenticated account). At least one must be present. When both are sent on subscribe, the server links them to one user record.

Field	Type	Usage
deviceId	string	Anonymous identity — subscription / pre-login
userId	string	Firebase or auth UID — logged-in user

3. Conversation flows

3.1 Flow A — Role Play (structured scenario)

Step	Action	Endpoint
1	Load conversation types	GET /v1/conversation-types
2	Load topics for selected type	GET /v1/conversation-types/{slug}/topics
3	Create chat (prompt stored server-side)	POST /v1/chats mode=role_play
4	Send user turn	POST /v1/chats/{chatId}/messages
5	Load history (optional)	GET /v1/chats/{chatId}/messages

Required POST /v1/chats fields: mode, learningLanguageName, typeId, topicId. Optional: difficultyKey (dl_beginner | dl_intermediate | dl_advanced).

3.2 Flow B — Free Chat (AI Conversation)

Step	Action	Endpoint
1	Create chat directly (no topic)	POST /v1/chats mode=free_chat
2	Send user turn	POST /v1/chats/{chatId}/messages
3	Load history (optional)	GET /v1/chats/{chatId}/messages

Do not send topicId with mode free_chat. Do not call the topics endpoint for ai-conversation.

3.3 Mode comparison

mode	UI entry	typId	topicId	Chat title
role_play	Type, Topic, Chat	Required	Required	Topic title
free_chat	AI Conversation	Ignored	Must omit	AI Conversation

4. Endpoint specification

4.1. Health check

Returns service and database connectivity status. No authentication.

Property	Value
Method	GET
URL	https://ailanguage.recipehubapi.com/health
Authentication	None

Response

```
{
  "ok": true,
  "service": "lingua-ai-api",
  "mongo": true
}
```

4.2. Subscribe user

Upsert user after in-app purchase or first launch.

Property	Value
Method	POST
URL	https://ailanguage.recipehubapi.com/v1/users/subscribe
Authentication	x-api-key required

Request body

```
{
  "deviceId": "string",
  "userId": "string (optional)",
  "subscriptionActive": true,
  "subscriptionProvider": "google"
}
```

Response

```
{
  "user": {
    "_id": "ObjectId",
    "deviceId": "string",
    "userId": "string | null",
    "subscriptionActive": true,
    "lastSeenAt": "ISO8601"
  }
}
```

4.3. List conversation types

Returns 9 active types with topicsCount. Optional query: filter=all|daily_life|work|travel.

Property	Value
Method	GET
URL	https://aialanguage.recipehubapi.com/v1/conversation-types
Authentication	x-api-key required

Response

```
{
  "types": [
    {
      "_id": "ObjectId",
      "slug": "daily-life",
      "title": "Daily Life",
      "topicsCount": 18
    }
  ]
}
```

4.4. List topics

Returns topics for a conversation type. Replace slug in path. Not available for ai-conversation (400).

Property	Value
Method	GET
URL	https://aialanguage.recipehubapi.com/v1/conversation-types/daily-life/topics
Authentication	x-api-key required

Response

```
{
  "type": { "_id": "6ab2292281d16f67ce31b18a", "slug": "daily-life" },
  "topics": [ { "_id": "6ab2292281d16f67ce31b195", "title": "At a Coffee Shop" } ]
}
```

Live IDs valid as of document date. Re-fetch after server re-seed.

4.5. Create chat — Role Play

Creates chat and stores systemInstruction server-side. Response excludes systemInstruction.

Property	Value
Method	POST
URL	https://aialanguage.recipehubapi.com/v1/chats
Authentication	x-api-key required

Request body

```
{
  "deviceId": "client-device-id",
  "mode": "role_play",
  "learningLanguageName": "Spanish",
  "typeId": "6ab2292281d16f67ce31b18a",
  "topicId": "6ab2292281d16f67ce31b195",
  "difficultyKey": "dl_beginner"
}
```

Response

```
HTTP 201
{
  "chat": {
    "id": "ObjectId",
    "mode": "role_play",
    "title": "At a Coffee Shop"
  }
}
```

400 if topicId missing or topic does not belong to typeId.

4.6. Create chat — Free Chat

Direct tutor session. Server assigns ai-conversation type internally.

Property	Value
Method	POST
URL	https://ailanguage.recipehubapi.com/v1/chats
Authentication	x-api-key required

Request body

```
{
  "deviceId": "client-device-id",
  "mode": "free_chat",
  "learningLanguageName": "Spanish"
}
```

Response

```
HTTP 201
{
  "chat": {
    "id": "ObjectId",
    "mode": "free_chat",
    "title": "AI Conversation"
  }
}
```

4.7. Send message

Submits one user turn. Server loads prompt and history from DB, calls Groq, persists messages on success.

Property	Value
Method	POST
URL	<code>https://aialanguage.recipehubapi.com/v1/chats/{chatId}/messages</code>
Authentication	x-api-key required

Request body

```
{
  "deviceId": "client-device-id",
  "userMessage": "Hola, quiero un cafe por favor."
}
```

Response

```
HTTP 200
{ "text": "assistant reply string" }

-- or on Groq failure --
{ "text": null }
```

Do NOT send systemInstruction or history. userMessage max 4000 characters.

4.8. Get messages

Returns ordered message history for the chat.

Property	Value
Method	GET
URL	<code>https://aialanguage.recipehubapi.com/v1/chats/{chatId}/messages?deviceId=client-device-id</code>
Authentication	x-api-key required

Response

```
{
  "chatId": "ObjectId",
  "title": "string",
  "mode": "role_play",
  "messages": [
    { "role": "user", "text": "string", "createdAt": "ISO8601" },
    { "role": "model", "text": "string", "createdAt": "ISO8601" }
  ]
}
```

4.9. List chats

Returns chats for owner sorted by lastMessageAt descending.

Property	Value
Method	GET
URL	https://aialanguage.recipehubapi.com/v1/chats?deviceId=client-device-id
Authentication	x-api-key required

Response

```
{
  "chats": [
    { "_id": "ObjectId", "title": "string", "mode": "role_play", "lastMessageAt": "ISO8601" }
  ]
}
```

systemInstruction is never included in responses.

4.10. Delete chat

Deletes chat and all associated messages.

Property	Value
Method	DELETE
URL	https://aialanguage.recipehubapi.com/v1/chats/{chatId}?deviceId=client-device-id
Authentication	x-api-key required

Response

```
{ "deleted": true }
```

5. Client implementation rules

Rule	Requirement
Prompt handling	Never send systemInstruction on message requests
History handling	Never send message history — server loads from DB
Chat creation	POST /v1/chats once per session; store returned chat.id
Identity	Same deviceId or userId on create, message, list, delete
Null AI response	Handle { "text": null } without crash
HTTPS	Production base URL must use TLS
Key storage	Store x-api-key securely; do not log in production

6. Verification commands

```
curl -s https://ailanguage.recipehubapi.com/health

curl -s https://ailanguage.recipehubapi.com/v1/conversation-types \
-H "x-api-key: KJSD84JIDNMZI83_3USLA983@8347SLDKJLE21D"

curl -s https://ailanguage.recipehubapi.com/v1/conversation-types/daily-life/topics \
-H "x-api-key: KJSD84JIDNMZI83_3USLA983@8347SLDKJLE21D"

# Swagger UI: https://ailanguage.recipehubapi.com/docs
```

End of document. Contact backend team before modifying client contracts.